VISHWAS · SIH26079 · TEAM HELLFIRE LABS

Dhruv

Users, impact and sourcing

Final prep — script, judge Q&A, and every number traced

BBA / BA | Slide 5 (second half) · Slide 6 (first half) | about 1 minute 15 of speaking, in two places

YOUR JOB

You close the argument and you own provenance. The single most valuable thing you can say in Q&A is why a number is NOT on the deck — that is what makes the rest of them credible.

1 RUNNING ORDER

Mahika leads. Your rows are highlighted.

#	SPEAKER	WHAT HAPPENS	TIME
1	Mahika	Slide 1 — opens. Team, problem statement, the hook.	~40s
2	Arnav	Slide 2 — what VISHWAS is, and why the forecast underneath decays.	~70s
3	Devansh	Slide 2 — the measurement, and the result.	~60s
4	Ankit	Slide 3 — opens it: what an ensemble is, and why we score against rain gauges.	~35s
5	Devansh	Slide 3 — the pipeline, the stack, features, label, model, the map.	~75s
6	Mahika	Bridge into slide 4.	~10s
7	Paras	Slide 4 — feasibility, the numbers, the three defects.	~65s
8	Arnav	Slide 4 — viability, the four risks, the precision choice.	~40s
9	Mahika	Slide 5 — the three interviews.	~45s
10	**Dhruv**	**Slide 5 — who it serves, MoES's numbers, the benefits. Then slide 6 — sources.**	**~75s**
11	Devansh	Slide 6 — our own measurement.	~20s
12	Mahika	Closes. The tagline, then into questions.	~20s

2 THE STORY — EVERYONE MUST BE ABLE TO SAY THIS

If a judge stops you in the corridor. Thirty seconds. Learn the shape, not the words.

> Every weather forecast India issues arrives with no honesty label.
>
> Forecasters judge confidence today by ensemble spread — run the model 11 times from slightly different starting conditions, and if the runs disagree, distrust the forecast. We measured that method against India's own rain gauges, across all four seasons and three years.
>
> Compared crudely it looks strong: ensemble spread appears to explain 43.9% of the variance in forecast error. Compared fairly — within the same rainfall conditions — it explains under 1%.
>
> So we built the replacement. On that same fair comparison it scores 0.565 and VISHWAS scores 0.728, on a test year the model had never seen.
>
> ***Most teams pitch a hypothesis. We pitch a measurement.***

3 WHAT CHANGED IN YOUR SCRIPT

Your wording is your own. Only these points moved.

- Your greeting is removed. Mahika greets for the team, and by Slide 5 a fresh “good evening” reads oddly — it also said a different time of day to hers.
- The closing “Thank you” is removed. It signals an ending, but Devansh takes the last card after you — and your line before it is his cue.
- One clause added about the false-alarm rate, matching what is printed on the slide.
- Nothing else. Every figure and every citation in your script checked out, including the Lok Sabha reference.

4 YOUR SCRIPT

Grey boxes are stage directions — do not say them. Lines tagged TRIM are safe to cut if you are running long.

SLIDE 5 — SECOND HALF

▸ *Mahika hands to you: “...who VISHWAS can serve, and the broader impact of addressing this gap.”*

> Thank you, Mahika.

> At the centre of VISHWAS is the duty forecaster at an operational centre — NCMRWF or IMD — the person assessing confidence in every forecast cycle.

> Aur yahin se VISHWAS ka real-world impact start hota hai.

> Because downstream, district authorities, agricultural advisories, power scheduling and aviation can all act on the same forecast.

> And the need isn't something we're assuming — it comes from the problem owner's own numbers.

▸ *PAUSE*

IMD's heavy-rainfall warning accuracy runs from 85 percent at Day 1 down to 58 percent at Day 5, according to the Ministry of Earth Sciences' answer to Lok Sabha Question 2953, December 2025. Its verification also reports a substantial false-alarm rate at these lead times.

▸ *PAUSE*

Simple words mein — jaise-jaise forecast ka lead time badhta hai, uski skill decline hoti hai. And that's exactly the window we score.

So, what changes with VISHWAS? Three things.

Operationally, confidence stops being judged by eye and becomes computed, consistent, reviewable, and measurable against observations.

Economically, fewer people and vehicles are committed to forecasts that were never reliable.

And socially, district authorities, farmers and rescue services can act on the same forecast — with an honest reliability label attached.

SLIDE 6 — FIRST HALF

▸ *Carry straight on. Do not pause for a slide change.*

Now, coming to our sources.

The forecast archive is NOAA's public GEFS version 12 reforecast bucket.

For the observed truth, we're using IMD's 0.25-degree gridded daily rainfall data, 1901 to 2024, from IMD Pune.

The reforecast dataset itself is documented by Guan and colleagues, 2022, in Monthly Weather Review.

For background and verification practice, we've used IMD's Monsoon Report and Annual Report, MoES, the Central Water Commission, and relevant peer-reviewed literature.

Aur sabse important point —

Every figure on this deck traces to one of these sources, or to our own logs.

▸ *STOP. Do not say thank you. Those last words are Devansh's cue and he takes the final card.*

5 EVERY NUMBER YOU SAY, AND WHERE IT COMES FROM

If a judge challenges a figure, give the source — not a defence.

CLAIM	SOURCE / WHAT IT REALLY MEANS
85% → 58%, Day 1 to Day 5	MoES, Lok Sabha Q. 2953, 17 Dec 2025. Full series: 85, 73, 67, 63, 58. Verified directly at moes.gov.in.
'a substantial false-alarm rate'	Deliberately not a number — see your Q&A. The IMD Monsoon Report 2024 supports that FAR is reported and substantial; it does not support a specific figure from us.
Guan et al. (2022), Mon. Wea. Rev. 150(3), 647–665	Verified through AMS. This is the correct citation for the GEFSv12 reforecast dataset — an earlier draft miscited it.
All government or peer-reviewed	IMD, MoES, CWC, NOAA; and QJRMS, Met. Apps, Mon. Wea. Rev., Risk Analysis, Agricultural Systems.

6 QUESTIONS THE JUDGES WILL ASK YOU

Yours to answer. Read them out loud once — an answer you have said aloud comes back under pressure; one you have only read does not.

Q1 Where does 85% to 58% come from?

The Ministry of Earth Sciences' own answer to Lok Sabha Question 2953, 17 December 2025. The full series is 85, 73, 67, 63 and 58 percent for Day 1 through Day 5. It is a PDF on moes.gov.in.

It also reports about a 40% improvement on 2014, which we did not put on the slide because it argues the other way and we did not want to quote only the half that helps us.

Q2 You say 'a substantial false-alarm rate'. How substantial?

I would rather not put a number on it, and there is a reason worth hearing.

We had a specific false-alarm ratio in an earlier draft. When we went to source it against the IMD Monsoon Report 2024, we could not put our hands on the table it came from — so we took it off.

The report does support that false-alarm ratio, probability of detection and threat score are reported, and that the rate is substantial. It does not support that specific pair of numbers coming from us. Every other figure on this deck traces, and I would rather lose one number than weaken all of them.

⚠ This is your strongest answer in the room — use it if you get any opening. But do NOT say the figure itself out loud. It is an untraceable number attributed to an IMD report, and you would be quoting it to judges from that ministry. The story works without it; the number is the one thing that can turn it against you.

Q3 What is the economic impact in rupees?

We do not have a figure we can defend, so we do not give one.

Our earlier drafts had damage totals and cost savings and we removed all of them — partly because they measured flood damage in general rather than forecast reliability specifically, and partly because under a Smart Automation framing they would be measuring the wrong thing.

What we can state is the mechanism: fewer mobilisations for rain that does not arrive, and less forecaster time spent on a judgement that currently has no evidence behind it.

Q4 Who is the actual customer?

The operational centre. NCMRWF or IMD would run this as a layer over a forecast they already produce — it is not a separate product a district officer buys.

The district officer is the beneficiary. The deployment point is the centre.

Q5 How is this different from just making the forecast better?

It is a different thing entirely. Improving the forecast is a decades-long modelling problem with a whole ministry working on it.

This tells you when today's forecast should not be trusted, using information that already exists in the ensemble run that has already been done. We detect busts. We do not correct forecasts.

Q6 **Are all your sources government or peer-reviewed?**

Yes. Government: IMD, the Ministry of Earth Sciences, the Central Water Commission, and NOAA for the forecast archive.

Peer-reviewed: Guan and colleagues 2022 in Monthly Weather Review, Ashrit and colleagues 2020 from NCMRWF in Meteorological Applications, Kumar and colleagues 2008 in the Quarterly Journal, LeClerc and Joslyn 2015 in Risk Analysis, and Kusunose and Mahmood 2016 in Agricultural Systems. Plus the WMO forecast verification standard for the metric definitions.

Q7 **What happens after the hackathon?**

Mid-September we extend from two years of training data to five — incremental, no re-downloading, no new code.

If we are selected: October to November is the operational path on live forecasts, a forecaster-facing confidence map, and a real Indian bust case study. Then the Grand Finale in December.

Q8 **The problem statement asks for a prototype dashboard or API. Where is it?**

Not built yet, and I would not claim otherwise. What exists today is the confidence map you saw on slide 3, generated per lead day, and a persisted model artifact that reproduces its predictions exactly — so the output layer is real, the interface around it is not.

That is the October to November work on our roadmap, and it is a wrapper around a system that already produces the output rather than new science.

⚠ The PS names five expected outcomes: confidence map, bust probability, error-prone area detection, explainable output, and a prototype dashboard/API. We have the first four. Do not let a judge discover the fifth gap before you name it.

Q9 **Why should MoES fund this rather than more compute for the forecast model?**

Because they are not alternatives at the same price. More compute improves the forecast marginally and expensively. This runs on CPU as a layer over a run that has already happened, and it tells a forecaster which of today's outputs to distrust.

And on our own measurement, the method they use for that judgement today explains under 1% of the variance in forecast error once you compare like with like. That gap is available now.

7 IF IT IS NOT YOURS, HAND IT OVER

Naming the right teammate reads as a team that knows its own work. Guessing reads as one that does not.

IF THEY ASK ABOUT…	SAY THE NAME
The model, the score, the method	Devansh
The data pipeline and the fetch	Devansh
What an ensemble is, why gauges rather than reanalysis	Ankit
A documented bust example	Arnav
Metrics, statistics, the defects	Paras
Viability, the risks, why GEFS and not NEPS	Arnav
What the interviewees said	Mahika

SITUATION	WHAT TO SAY
You don't know the answer	“I don't have that number to hand and I'd rather not guess — can I come back to you on it?” That is a strength for this team, not a weakness: when we could not trace one of our own figures, we removed it rather than present it.
It's a teammate's area	“That's <name>'s part — <name>, do you want to take that?” Say the name. Don't answer for them.
A judge challenges a number	Give the source, not a defence. Every figure on the deck traces to a government source, a peer-reviewed paper, or our own logs.
A judge says 'you're just predicting heavy rain'	Anyone can answer this one: we removed the rainfall forecast from the model entirely and the score went UP, 0.728 to 0.731.

8 NEVER SAY THIS

Every one of these was a real error caught during the build, or a rule set deliberately.

- × Never say '43.9% of forecast error'. It is 43.9% of the VARIANCE in forecast error. This is the single easiest thing to get wrong under pressure.
- × Never quote the margin as '+0.163'. Say '+0.13 to +0.16'.
- × Never say 'deciles' or 'the wettest tenth' about the model result. It is EIGHT bins. (Deciles are correct only for the separate Phase 1 correlation analysis.)
- × Never quote any accuracy figure. Base rate is about 6%, so accuracy is meaningless — and quoting it signals a team that does not understand imbalance.
- × Never say 'AI-adjusted forecast' or imply we correct the forecast. We DETECT busts. We do not correct anything.
- × Never volunteer the pooled ROC-AUC pair (0.909 vs 0.885). It is a Q&A-only answer if a judge asks directly what the crude comparison gives. Putting it forward would commit, in our own work, exactly the error our pitch accuses the incumbent of.
- × Never quote +0.663 as 'the baseline we beat', and never quote a margin range containing +0.035 (that one came from a filter conditioning on the outcome).
- × Never call the regions 'IMD official regions'. They are approximate lat/lon boxes.
- × Never present the confidence map's date as evidence of general performance. It is ONE date, illustrating the form of the product.
- × Never convert the three field interviews into percentages or 'N% of users said'. They are three exploratory conversations.
- × Never quote 98%→46%, ₹13,331 cr, '50% faster', '~30% improvement', or a 0.34–0.37 false-alarm ratio. All were in earlier drafts; none could be traced; all were removed. Do not say any of these numbers out loud even while explaining that you removed them.
- × Never say SHAP and permutation importance 'agree on the drivers'. They pick the same top three in a different order.

9 NUMBER CHEAT SHEET

Same in all six packs. If two of you quote a figure differently in the same room, that is the damage.

WHAT	NUMBER	WATCH OUT
Spread explains, pooled	43.9% of the VARIANCE in forecast error	not '43.9% of forecast error'
Spread explains, fair comparison	0.17%–0.90% → say 'under 1%'	never quote +0.041 alone
VISHWAS score (stratified ROC-AUC)	0.728	—

WHAT	NUMBER	WATCH OUT
Ensemble spread score	0.565	—
Margin	+0.13 to +0.16 · 95% CI +0.108 to +0.193	never '+0.163' alone
Matched 10% alert budget	VISHWAS catches 64% of busts vs 55%	—
POD / FAR / CSI / ETS at 10%	0.641 / 0.586 / 0.336 / 0.300 vs 0.546 / 0.647 / 0.273 / 0.235	—
Features	23, all from GEFS, none from the truth	not 24
Ensemble members	11 (control + 10 perturbed)	—
Domain	1,252 of 5,041 gridpoints (24.8%), land only	—
Data fetched	50.3 GB · 104 forecast dates	—
Labelled rows	1,302,080 total · 651,040 in the test set	don't conflate the two
Train / test	trained on 2018 · tested on all 52 Wednesdays of 2019	—
Base rate	about 6%	so never quote accuracy
Bust definition	|forecast − observed| above the 95th percentile, per region and lead time, threshold set on training data only	—
Rainfall bins	EIGHT bins · 40.92% bust rate in the wettest, 0.15% in the driest	never 'deciles' or 'wettest tenth'
Ablations	full 0.728 · no month 0.722 · no rainfall forecast 0.731 · no 850 shear 0.726 · spread-only 0.562	—
Underlying GEFS, heavy rain (≥64.5 mm)	detects 23% of events at Day 1, 0.2% at Day 10	this is the forecast, NOT our model
IMD warning accuracy (MoES)	85, 73, 67, 63, 58% for Day 1–5	Lok Sabha Q.2953, 17 Dec 2025
The tie	ensemble spread is exactly 0.00 mm on 189,758 test rows — a third of the set	—
Calibration	Brier 0.04217 vs 0.04749 · resolution 1.4× the incumbent's	—

ONE LAST THING

The strongest thing about this project is not the score. It is that we measured the method used today before proposing a replacement, and that we removed our own figures when we could not source them. If you land one impression, land that one.